RÉSUMÉ · 2026 · NAIROBI, KENYA



Maureen Njoki

Software & Data Engineer · Geospatial Systems

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Currently leading the NOVA digital presence redevelopment for Spowdi Africa.

PROFILE

Engineer and data scientist with 7+ years across national-scale digital systems, applied machine learning and web platforms. I build the whole stack — Django and PostgreSQL backends, payment integrations, mobile-first interfaces — and I bring something most web teams don't: a geospatial engineering background, so I understand land, counties, crops and satellite-derived field data from the inside. I work in small, tested increments, and I write documentation a non-technical team can actually maintain a site with.

SPECIALIST PRACTICE

Geospatial Engineering

- Led digitisation of the **Kenya National Land Information System** — national-scale spatial data migration into SQL Server and PostgreSQL/PostGIS, delivered to Cabinet Secretary level.
- Engineered migration scripts loading satellite, meteorological and relational datasets into a unified spatial store.
- Authored the SOPs and validation procedures adopted by a 600+ person team, then trained and onboarded 1,000+ personnel.
- Topographical and cadastral surveying, field data capture and feature digitisation (2016–2018).
- Tooling: ArcGIS, QGIS, Google Earth Engine, PostGIS, satellite imagery analysis, GPS field mapping.
- Author of *Spatial Tidbits* — a technical publication on spatial SQL, data engineering and query optimisation.

Data Science

- Built an **IFRS 9 Expected Credit Loss model** and credit-scoring models for Kenyan SACCOs, with documented methodology and model review.
- Designed automated ETL pipelines (SQL, Apache NiFi, PySpark) with explicit data-quality checks, so outputs stay auditable for compliance review.
- Delivered an end-to-end sentiment-analysis pipeline: Facebook API → SQL Server → ML model → BI dashboard, for a communications department.
- Approved BI reporting for credit operations; built dashboards that let business teams answer their own questions.
- Tooling: SQL/T-SQL, SQL Server, PostgreSQL, PySpark, Databricks, Power BI, Scikit-Learn.

AI Engineering

- Fine-tuned **YOLOv8** and **Vision Transformer** architectures for object detection and classification across domains.
- Competed on the **NASA Harvest Field Boundary Detection** and INEGI UN-GGIM Human Settlement Detection challenges, plus malaria-cell detection.
- Builds interpretable models — SHAP attribution and audit-ready documentation — so outputs can be explained to decision-makers.
- Serves models in production behind Django and FastAPI services, with a focus on data integrity and correctness.
- MLOps practice: model versioning, monitoring, evaluation discipline, hyperparameter tuning and data augmentation.
- Tooling: Python, PyTorch, Keras, TensorFlow, Scikit-Learn, SHAP.

FIT FOR THE NOVA REDEVELOPMENT

WHAT THE ENGAGEMENT NEEDS	WHAT I BRING
Mobile-first, plain-language pages for farmers	Built a public subscription platform mobile-first and verified it by headless measurement at 320, 360, 390, 768 and 1280 px — no horizontal overflow at any width.
Audience-specific CTAs that route leads correctly	Designed and shipped server-validated public forms with database-backed records, per-record access tokens and audit trails.
Main site plus a separate ke/ subdomain	Multi-tenant Django architecture with per-tenant scoping — one codebase, isolated data, separate public surface.
A CMS the team can update without a developer	Role-based Django admin with read-only enforcement on financial records; publishing on WordPress since 2021 (Spatial Tidbits).
Agricultural and land context	Digitised the Kenya National Land Information System, and trained models to delineate field boundaries from satellite imagery — land size, crop type and county are data I have worked with, not just form fields.

WHAT THE ENGAGEMENT NEEDS	WHAT I BRING
Delivery you can hold to a schedule	Product owner and QA lead on a national geospatial system: test strategy, release coordination, weekly stakeholder reporting.

SELECTED SOFTWARE ENGINEERING

BoardLot Africa — subscription & payments platform

DJANGO 5 · POSTGRESQL · CLOUD9 / M-PESA

A membership platform for a reader-supported publication: visitors choose a tier, pay by M-Pesa STK Push, and access is granted automatically. Built end to end — data model, payment integration, admin, responsive front end, test suite and deployment configuration.

- **Three independent locks on the payment callback.** A per-record token in the URL path, a constant-time comparison of an HMAC header secret, and a provider re-fetch that must independently confirm success and amount before a record is marked paid. Each fails closed.
- **Idempotent by construction.** A provider retry cannot double-charge or grant a second year; duplicate callbacks resolve to a no-op and the row lock serialises concurrent delivery.
- **Append-only financial ledger.** Enforced at four levels — model `save()`, `delete()`, the queryset layer and a database constraint — so a correction is a new compensating entry, never an edit. Financial history cannot be rewritten, by anyone, including me.
- **Transactional integrity.** Membership activation happens inside the same transaction as the payment confirmation, so a failure rolls money back rather than recording a payment against no access. Outbound email is deferred to `on_commit`: a broken mailer cannot un-pay a reader.
- **Verified, not asserted.** 157 automated tests, plus production settings that refuse to boot without secrets, and layout verified by DevTools Protocol measurement rather than eyeballed screenshots.

National geospatial information management system

PRODUCT · QA · DELIVERY

Product owner and QA lead for a government-facing spatial system. Introduced Cypress test automation and a regression suite, owned release coordination and production triage, and acted as the delivery liaison between engineering and ministry stakeholders.

Risk scoring services for SACCOs

PYTHON · DJANGO · FASTAPI · SQL

Python services that ingest client data and serve credit-risk models, built around data integrity and correctness: validated ingestion, documented methodology, and model review that stands up to a compliance conversation.

PROFESSIONAL EXPERIENCE

Data Scientist & Data Engineer

Feb 2026 – Present

Zealtech Data Solutions · Nairobi, Kenya

- Built IFRS 9 Expected Credit Loss and credit-scoring models for Kenyan SACCOs, with documented methodology and model review.
- Developed Python (Django, FastAPI) services to ingest client data and serve risk-scoring models, prioritising data integrity and correctness.
- Built automated ETL pipelines (SQL, Apache NiFi, PySpark) with documented data-quality checks, keeping the system auditable for compliance review.
- Delivered an end-to-end sentiment-analysis pipeline for a communications department and approved BI reporting for credit operations.

Software QA Lead & Product Manager

Feb 2022 – Jul 2024

Geo-Project Kenya · Kenya

- Owned the product lifecycle of a national geospatial information management system: feature planning, test strategy, release coordination and production triage.
- Led a Cypress test-automation strategy and regression suite; reviewed designs and test plans with engineering, product and business stakeholders.
- Acted as delivery liaison between technical teams and government and business stakeholders, tracking issues from triage through resolution.

GIS Analyst & Technical Lead

Nov 2019 – Feb 2022

Geo-Project Kenya · Kenya

- Led digitisation of the Kenya National Land Information System, engineering migration scripts to load satellite, meteorological and structured datasets into SQL Server and PostgreSQL/PostGIS.
- Developed and documented SOPs adopted across a 600+ person team; established validation procedures that fed Cabinet Secretary-level reporting.
- Trained and onboarded 1,000+ personnel on new tools and workflows, managing change communications and measuring adoption.

ML Challenge Competitor

Feb 2022 – Present

Zindi · Solafune · international platforms

- Applied hyperparameter tuning, data augmentation and rigorous model evaluation across independent applied ML challenges, including NASA Harvest and INEGI UN-GGIM.

Survey Assistant

Nov 2016 – Jul 2018

Ramani Land Services & Highland Surveyors · Kenya

- Conducted topographical and cadastral surveys and digitised land features to support infrastructure planning and land-use decisions.

TECHNICAL SKILLS

LANGUAGES	Python · SQL / T-SQL · JavaScript · HTML & CSS
WEB & BACKEND	Django · FastAPI · PostgreSQL · SQL Server · PostGIS · REST APIs · payment integrations (M-Pesa STK Push) · mobile-first responsive CSS · Git & CI gates
DATA & ML	Scikit-Learn · PyTorch · Keras · TensorFlow · SHAP · PySpark · Databricks · Apache NiFi · Power BI
GEOSPATIAL	ArcGIS · QGIS · Google Earth Engine · PostGIS · satellite imagery analysis · GPS field mapping · cadastral & topographical survey
PLATFORMS & CMS	WordPress (content operations & site administration) · Django admin (role-based, audited)
QUALITY	Cypress test automation · regression suite design · CI/CD · data-quality controls · audit-ready documentation

EDUCATION

MSc Computer Science — Computational Intelligence

2022 – Present

University of Nairobi

BSc Geospatial Engineering

2014 – 2019

University of Nairobi

CERTIFICATIONS & TRAINING

- Coursera Deep Learning Specialization — Google Developers ML Sub-Saharan Africa Bootcamp
- Kaggle — Geospatial Analysis
- Kaggle — Introduction to Machine Learning
- Coursera — Introduction to Machine Learning

COMMUNITY & LEADERSHIP

- **Co-Lead, Women in Machine Learning and Data Science (WiMLDS)** — Nairobi chapter.
- **Trainer, CodeYetu / KidsThatCode** — teaching ML and web development to youth aged 10–17 (Oct 2023 – Present).
- **Vice Curator, Global Shapers Nairobi Hub** — WEF-affiliated community impact projects in tech and environment (2022 – 2024).
- **President Elect & Admin Lead, Rotaract Club of Nairobi Thika Road** (2022 – 2024).
- **Mentor, Glow Foundation** — career and higher-education mentorship for girls (2022 – 2024).

CONTACT

Happy to walk through any of the above, or share the code. Maureennjoki6@gmail.com · +254 720 287 808 · LinkedIn